

PENCOR 4 mg Tablet

ANTIHYPERTENSIVE/BENIGN PROSTATIC HYPERPLASIA THERAPY AGENT

Each tablet contains:

Doxazosin mesylate eq. to Doxazosin 4 mg✧ **PRODUCT DESCRIPTION:**

White, square, biconvex tablet with engraved PENCOR 4 on one side and scored on the other

✧ **MECHANISM OF ACTION:****Pharmacology**

Doxazosin has a selective postsynaptic α_1 -adrenergic blocking action, which is thought to account primarily for its effects. **Hypertension:** Blockade of α_1 -adrenergic receptors by Doxazosin results in peripheral vasodilation, which produces a fall in blood pressure because of decreased peripheral vascular resistance.

Benign prostatic hyperplasia: Relaxation of smooth muscle in the bladder neck, prostate, and prostate capsule produced by α_1 -adrenergic blockade results in reduction in urethral resistance and pressure, bladder outlet resistance, and urinary symptoms. Doxazosin slightly lowers the levels of total cholesterol, low-density lipoprotein (LDL) cholesterol, and triglycerides. In addition, Doxazosin slightly increases high-density lipoprotein (HDL) cholesterol and the HDL/ total cholesterol ratio. These lipid effects appear to be result of Doxazosin's effect on lipid metabolism (i.e., increasing LDL receptor activity, decreasing intracellular LDL cholesterol synthesis, decreasing synthesis and secretion of very low-density lipoprotein [VLDL] cholesterol, stimulation of lipoprotein lipase activity, and decreasing the rate of cholesterol absorption). However, the implications of these changes are unclear.

Pharmacokinetics

Absorption: Well-absorbed from gastrointestinal tract; bioavailability is about 65%.

Protein binding: Very high (98 to 99%).

Biotransformation: Metabolized extensively in the liver. Although several active and inactive metabolites have been identified (2-piperazinyl, 6' and 7'-hydroxy, 6' and 7'-O-desmethyl, and 2-amino), there is no evidence that they are present in substantial amounts.

Half-life: Elimination- 19 to 22 hours; does not appear to be significantly influenced by age or mild to moderate renal impairment. Onset of action: Hypertension- 1 to 2 hours; there is a slight initial fall in blood pressure within the first hour, but the main hypotensive effect is apparent from 2 hours onwards. Benign prostatic hyperplasia (BPH)- Within 1 to 2 weeks.

Time to peak concentration: 1.5 to 3.6 hours.

Peak serum concentration: At steady state, there is a positive linear relationship between peak serum concentration and dose of Doxazosin. Following an oral dose of 1 mg Doxazosin, the standardized peak serum concentration was 9.6 mcg per L.

Time to peak effect: Antihypertensive- single dose: 2 to 6 hours.

Duration of action: Antihypertensive- Single dose: 24 hours.

Elimination: Fecal - Unchanged drug, about 5%; metabolites, 63 to 65%. Renal- 9%. In dialysis- Doxazosin is not removed by hemodialysis.

✧ **INDICATION:**

Hypertension: PENCOR is indicated for the treatment of hypertension and can be used as the sole agent to control blood pressure in the majority of patients. In patients inadequately controlled on single antihypertensive therapy, PENCOR may be used in combination with thiazide diuretic, beta-adrenoceptor blocking agent, calcium antagonist or an angiotensin-converting enzyme inhibitor.

Benign prostatic hyperplasia: PENCOR is indicated for the treatment of urinary outflow obstruction and symptoms associated with benign prostatic hyperplasia (BPH). PENCOR may be used in BPH patients who are either hypertensive or normotensive.

✧ **DOSAGE AND ADMINISTRATION:**

Oral

Hypertension: PENCOR is used in a once daily regimen: the initial dose is 1mg, to minimize the potential for postural hypotension and/ or syncope.

Dosage may then be increased to 2 mg after an additional one or two weeks of therapy and thereafter, if necessary to 4 mg. The majority of patients who respond to PENCOR will do so at a dose of 4 mg or less. Dosage can be further increased if necessary to 8 mg or the maximum recommended dose of 16 mg.

Benign prostatic hyperplasia: The recommended initial dosage of PENCOR is 1 mg given once daily to minimize the potential for postural hypotension and/or syncope. Depending on the individual patient's urodynamics and BPH symptomatology dosage may then be increased to 2 mg and thereafter to 4 mg and up to the maximum recommended dose of 8 mg. The recommended titration interval is 1-2 weeks. The usual recommended dose is 2-4 mg daily.

Children

The safety and efficacy of PENCOR in children have not been established.

Elderly

Normal adult dosage.

Patients with renal impairment

Since there is no change in pharmacokinetics in patients with impaired renal function, the usual adult dose of PENCOR is recommended. PENCOR is not dialyzable.

Patients with hepatic impairment

There are only limited data in patients with liver impairment and on the effect of drugs known to influence hepatic metabolism (e.g. Cimetidine). As with any drug wholly metabolized by the liver, PENCOR should be administered with caution to patients with evidence of impaired liver function.

✧ **PRECAUTION:**

Postural Hypotension/ Syncope: As with all α -blockers, a very small percentage of patients have experienced postural hypotension evidenced by dizziness and weakness, or rarely loss of consciousness (syncope), particularly with the commencement of therapy. When instituting therapy with any effective α -blocker, the patient should be advised how to avoid symptoms resulting from postural hypotension and what measures to take should they develop. The patient should be cautioned to avoid situations where injury could result, should dizziness or weakness occur during the initiation of PENCOR therapy.

Use with PDE-5 Inhibitors: Concomitant administration of an α -blocker with a PDE-5 inhibitor should be used with caution as it may lead to symptomatic hypotension in some patients.

Impaired liver function: As with any drug wholly metabolized by the liver, PENCOR should be administered with caution to patients with evidence of impaired hepatic function.

✧ **PREGNANCY AND LACTATION:****Use during pregnancy**

Although no teratogenic effects were seen in animal testing, reduced fetal survival was observed in animals at extremely high doses. These doses were approximately 300 times the maximum recommended human dose. As there are no adequate and well-controlled studies in pregnant women, the safety of PENCOR during pregnancy has not yet been established. Accordingly, PENCOR should be used only when, in the opinion of the physician, the potential benefit outweighs the potential risk.

Product	Code No.	Dimension	Packaging type	Thickness
PENCOR 4	IPMY 0071	W. 15.5 x L 26.4 cm.	Wood Free Paper (กระดาษปลอดกาว)	60 g (0.08 mm.)
 PM Specification Designed by: (PDM, ASD, PCCA, PCCA, PCCA, PCCA) Checked by: (PDM/ASPD) Approved by: (MD, Only Patient and Cases) PLC: (Date) LRA: (Date) ICC-PM: (Date) CUSTOMER/SALES DEPT: (Date)				

Use during lactation

Contraindicated. See Contraindications.

✧ **SIDE EFFECT:**

Note: A "first-dose orthostatic hypotensive reaction" sometimes occurs with the initial dose of Doxazosin, especially when the patient is in the upright position. Syncope or other postural symptoms such as dizziness may occur. Subsequent occurrence with dosage increases is also possible. Incidence appears to be dose-related; thus it is important that therapy be initiated with the 1 mg dose.

Patients who are volume-depleted or sodium-restricted may be more sensitive to the orthostatic hypotension effects of Doxazosin, and the effect may be exaggerated after exercise.

Hypotensive side effects are more likely to occur in geriatric patients.

The following side/ adverse effects have been selected on the basis of their potential clinical significance (possible signs and symptoms in parentheses where appropriate) - not necessarily inclusive:

Those indicating need for medical attention

Incidence less frequent

Dizziness; Vertigo (dizziness or lightheadedness)

Incidence less frequent

Arrhythmias (irregular heartbeat); Dyspnea (shortness of breath); Orthostatic hypotension (dizziness or lightheadedness when getting up from a lying or sitting position; sudden fainting); Palpitations (pounding heartbeat); Peripheral edema (swelling of feet or lower legs); Tachycardia (fast heartbeat)

Incidence rare

Priapism (painful or prolonged erection of the penis)

Those indicating need for medical attention only if they continue or are bothersome

Incidence more frequent

Headache; Unusual tiredness

Incidence less frequent

Nausea; Nervousness, restlessness, or unusual irritability; Rhinitis (runny nose); Somnolence (sleepiness or unusual drowsiness)

A. Benign Prostatic Hyperplasia

The incidence of adverse events has been ascertained from worldwide clinical trials in 965 BPH patients. The incidence rates are based on combined data from seven placebo-controlled trials involving once daily administration of PENCOR (Doxazosin mesylate) in doses of 1-16 mg in hypertensives and 0.5-8 mg in normotensives. The adverse events when the incidence in the PENCOR group was at least 1%. No significant difference in the incidence of adverse events compared to placebo was seen except for dizziness, fatigue, hypotension, edema and dyspnea. Dizziness and dyspnea appeared to be dose-related.

✧ **CONTRAINDICATION:**

Doxazosin is contraindicated in patients with a known hypersensitivity to Quinolones, Doxazosin, or any ingredients of the products.

✧ **DRUG INTERACTION:**

Doxazosin is highly bound to plasma proteins (98%). *In vitro* data in human plasma indicates that Doxazosin has no effect on protein binding of the drugs tested (Digoxin, Phenytoin, Warfarin or Indomethacin). No adverse drug interactions have been observed with thiazide diuretics, frusemide, beta-blocking agents, non-steroidal anti-inflammatory drugs, antibiotics, oral hypoglycemic drugs, uricosuric agents, or anticoagulants.

Concomitant administration of an alpha-blocker with a PDE-5 inhibitor may lead to symptomatic hypotension in some patients.

✧ **OVERDOSE AND TREATMENT:****Symptoms**

Circulatory failure

Treatment

Place the patient in the supine position and elevating the legs is most important; if shock is present, additional measures are necessary. Volume expanders may be used to treat shock, followed by administration of a vasopressor.

Symptomatic, supportive treatment and monitoring of fluid and electrolyte status.

Experience with PENCOR overdosage is limited. Two adolescents who each intentionally ingested 40 mg PENCOR with Diclofenac or Paracetamol, were treated with gastric lavage with activated charcoal and made full recoveries. A two-year-old child who accidentally ingested 4 mg PENCOR was treated with gastric lavage and remained normotensive during the five-hour emergency room observation period. A six-month-old child accidentally received a crushed 1 mg tablet of PENCOR and was reported to have been drowsy. A 32-year-old female with chronic renal failure, epilepsy and depression intentionally ingested 60 mg PENCOR (blood level 0.9 µg/mL; normal values in hypertensives = 0.02 µg/mL); death was attributed to a grand mal seizure resulting from hypotension. A 39-year-old female who ingested 70 mg PENCOR, alcohol and Dalmane (Flurazepam) developed hypotension which responded to fluid therapy. The oral LD₅₀ of Doxazosin is greater than 1,000 mg/kg in mice and rats. The most likely manifestation of overdosage would be hypotension, for which the usual treatment would be intravenous infusion of fluid. As Doxazosin is highly protein bound, dialysis would not be indicated.

✧ **STORAGE:**

Store at a temperature of not more than 30°C.

✧ **DOSAGE FORM AND PACKAGING AVAILABLE:**

4 mg Tablet, Blister 10x10's

✧ **DATE OF REVISION:**

May 24, 2023

Manufactured by:

UNISON LABORATORIES CO., LTD.

39 Moo 4, Klong Udomcholjorn, Muang Chachoengsao,

Chachoengsao 24000 Thailand

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